

AD EXP-8

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Batch: S-5

Code:

JAVA CODE:

MainActivity.java:

```
package com.example.mediaplayerapp;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.os.Handler;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.SeekBar;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    MediaPlayer mediaPlayer;
    ImageButton playPauseButton, nextButton, previousButton;
    SeekBar seekBar;
    TextView songTitle, artistName;
    ImageView albumArt;

    ArrayList<Song> playlist;
    int currentIndex = 0;
    Handler handler = new Handler();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        playPauseButton = findViewById(R.id.playPauseButton);
        nextButton = findViewById(R.id.nextButton);
        previousButton = findViewById(R.id.previousButton);
        seekBar = findViewById(R.id.songProgressSeekBar);
        songTitle = findViewById(R.id.songTitleTextView);
        artistName = findViewById(R.id.artistNameTextView);
        albumArt = findViewById(R.id.albumArtImageView);

        playlist = new ArrayList<>();
```

```

playlist.add(new Song("Perfect", "Ed Sheeran", R.raw.perfect, R.drawable.img1));
playlist.add(new Song("Yellow", "Coldplay", R.raw.yellow, R.drawable.img2));

setupSong(currentIndex);

playPauseButton.setOnClickListener(v -> {
    if(mediaPlayer.isPlaying()){
        mediaPlayer.pause();
        playPauseButton.setImageResource(R.drawable.ic_play);
    } else {
        mediaPlayer.start();
        playPauseButton.setImageResource(R.drawable.ic_pause);
    }
});

nextButton.setOnClickListener(v -> {
    currentIndex = (currentIndex + 1) % playlist.size();
    setupSong(currentIndex);
});

previousButton.setOnClickListener(v -> {
    currentIndex = (currentIndex - 1 + playlist.size()) % playlist.size();
    setupSong(currentIndex);
});
}

private void setupSong(int index){
    if(mediaPlayer != null){
        mediaPlayer.release();
    }

    Song song = playlist.get(index);
    mediaPlayer = MediaPlayer.create(this, song.getAudioResId());

    songTitle.setText(song.getTitle());
    artistName.setText(song.getArtist());
    albumArt.setImageResource(song.getImageResId());

    seekBar.setMax(mediaPlayer.getDuration());

    mediaPlayer.start();
    playPauseButton.setImageResource(R.drawable.ic_pause);

    updateSeekBar();
}

private void updateSeekBar(){
    handler.postDelayed(new Runnable() {
        @Override
        public void run() {

```

```

        if(mediaPlayer != null){
            seekBar.setProgress(mediaPlayer.getCurrentPosition());
            handler.postDelayed(this, 1000);
        }
    },1000);
}

@Override
protected void onDestroy() {
    super.onDestroy();
    if(mediaPlayer != null){
        mediaPlayer.release();
    }
}
}

```

Song.java:

```

package com.example.mediaplayerapp;

public class Song {

    private String title;
    private String artist;
    private int audioResId;
    private int imageResId;

    public Song(String title, String artist, int audioResId, int imageResId){
        this.title = title;
        this.artist = artist;
        this.audioResId = audioResId;
        this.imageResId = imageResId;
    }

    public String getTitle(){ return title; }
    public String getArtist(){ return artist; }
    public int getAudioResId(){ return audioResId; }
    public int getImageResId(){ return imageResId; }
}

```

XML CODE:

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```

android:background="#121212">

```
<ImageView
    android:id="@+id/albumArtImageView"
    android:layout_width="300dp"
    android:layout_height="300dp"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="80dp"
    android:scaleType="centerCrop"/>
```

```
<TextView
    android:id="@+id/songTitleTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/albumArtImageView"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="20dp"
    android:textColor="#FFFFFF"
    android:textSize="20sp"
    android:textStyle="bold"/>
```

```
<TextView
    android:id="@+id/artistNameTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/songTitleTextView"
    android:layout_centerHorizontal="true"
    android:textColor="#B0B0B0"
    android:textSize="16sp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_alignParentBottom="true"
    android:orientation="vertical"
    android:layout_marginBottom="50dp">
```

```
<SeekBar
    android:id="@+id/songProgressSeekBar"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginHorizontal="30dp"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
    android:orientation="horizontal">
```

```
<ImageButton
```

```
android:id="@+id/previousButton"  
android:layout_width="60dp"  
android:layout_height="60dp"  
android:background="@null"  
android:src="@drawable/ic_previous"/>
```

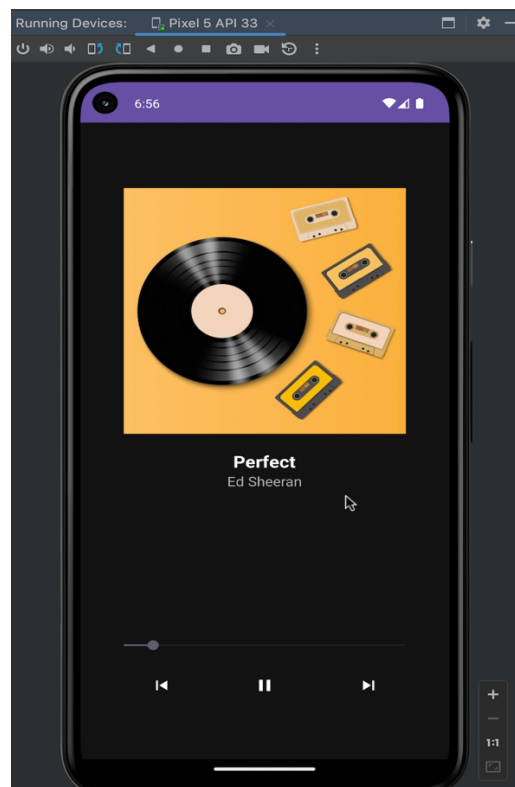
```
<ImageButton  
    android:id="@+id/playPauseButton"  
    android:layout_width="80dp"  
    android:layout_height="80dp"  
    android:background="@null"  
    android:src="@drawable/ic_play"  
    android:layout_marginHorizontal="40dp"/>
```

```
<ImageButton  
    android:id="@+id/nextButton"  
    android:layout_width="60dp"  
    android:layout_height="60dp"  
    android:background="@null"  
    android:src="@drawable/ic_next"/>
```

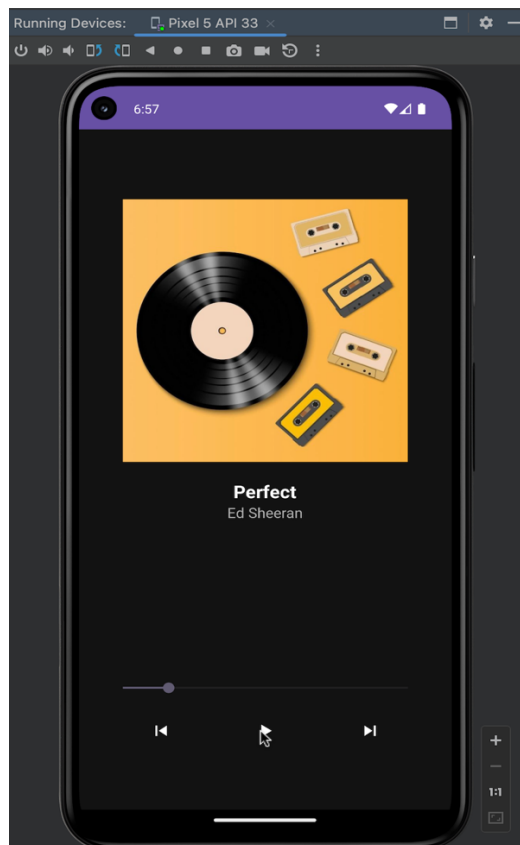
```
</LinearLayout>  
</LinearLayout>  
</RelativeLayout>
```

O/P:

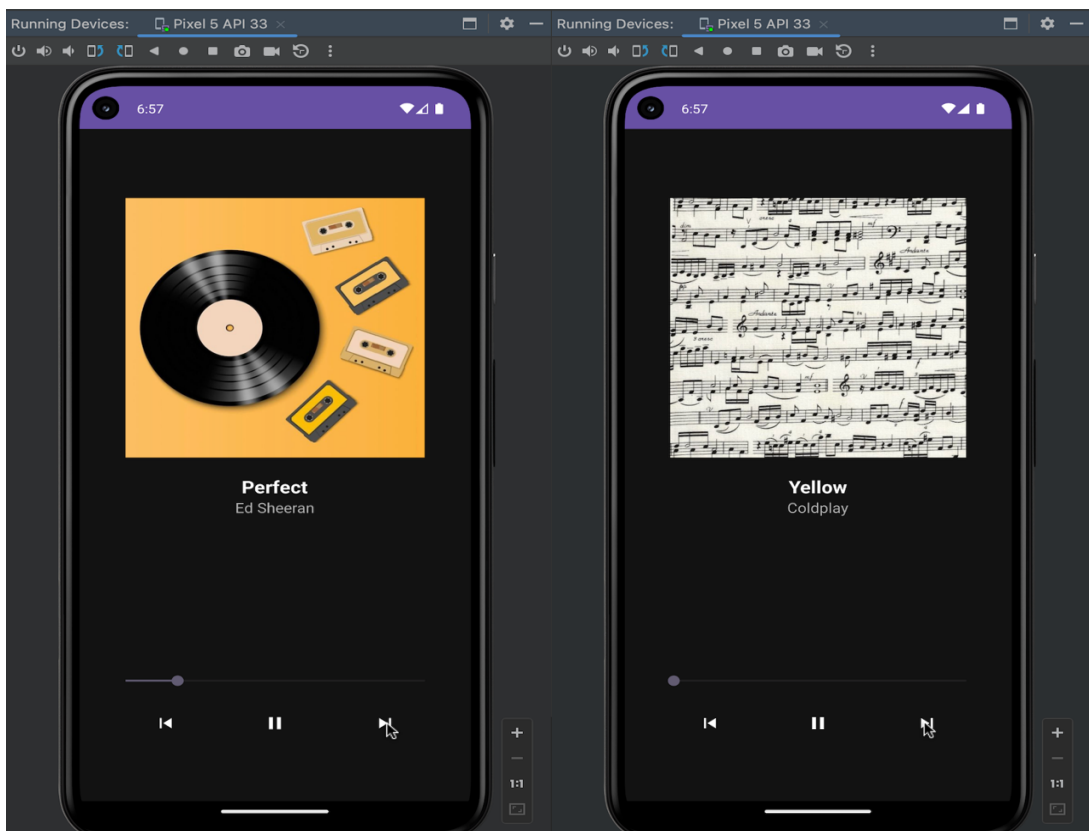
1.) Play



2.) Pause



3.) Forward



4.) Backward

